

Department of Electronics and Electrical Engineering
Indian Institute of Technology Guwahati

EE101 - Basic Electronics

Tutorial 6

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1. Select the values of the resistors R_1 and R_2 in the circuit shown in Figure 1 such that $v_R(0^+) = 10V$ and $v_R(1\text{ ms}) = 5V$.

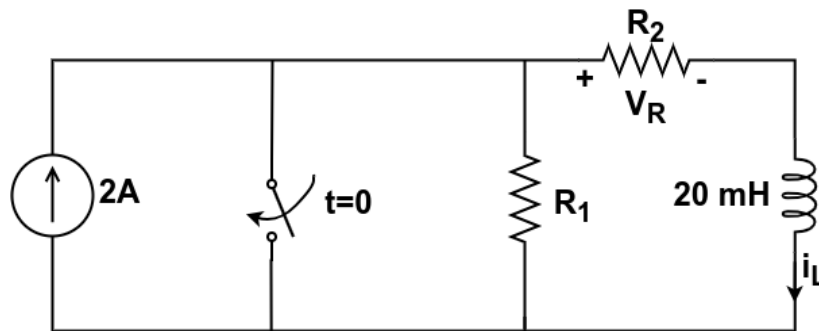
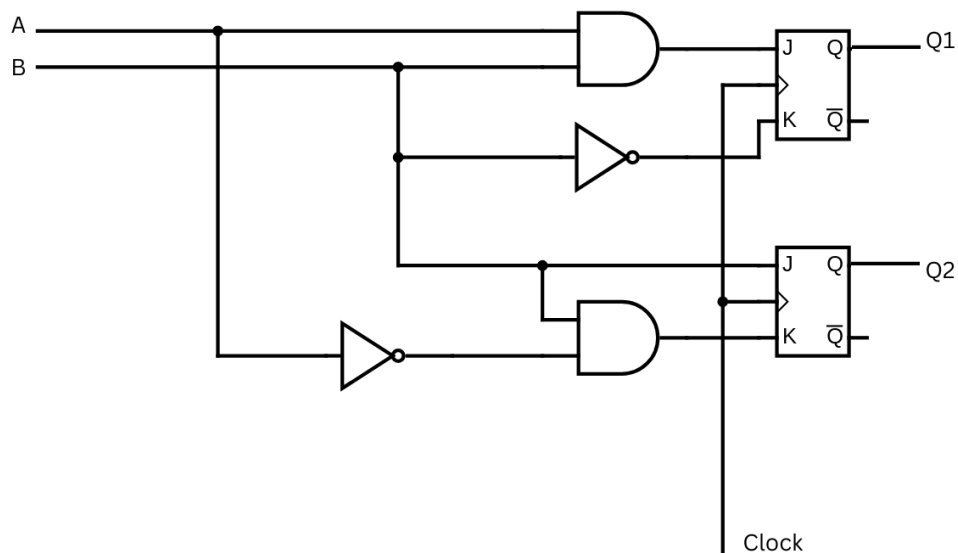
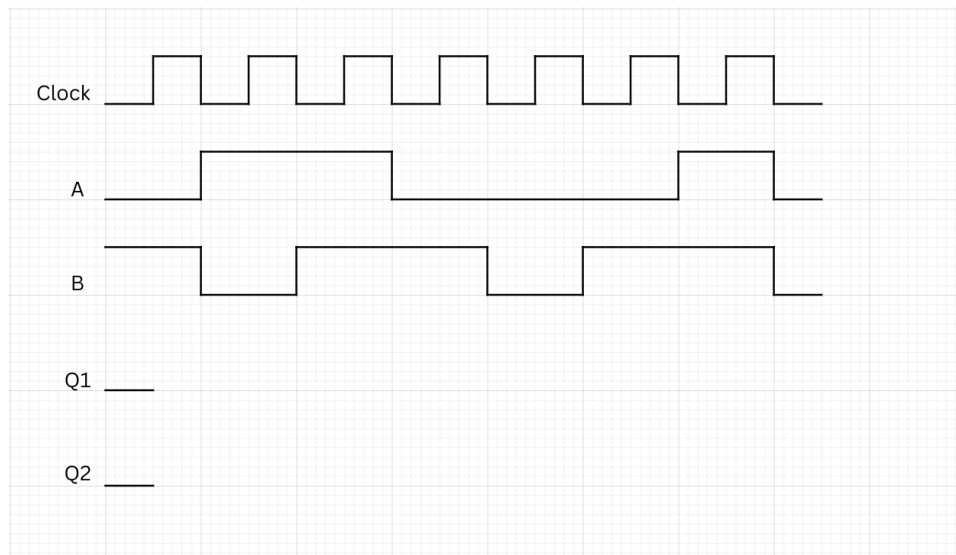


Figure 1 Electric circuit with an independent current source, a switch, resistors and an inductor. The switch closes at $t = 0$.

2. Construct a negative edge triggered JK Flip-flop using a positive edge triggered T Flip-flop and logic gates.
3. Consider the circuit shown in the figure below.



Plot $Q1$ and $Q2$ for the inputs shown below.



4. Consider the sequential circuit shown in the figure below, consisting of 4 D Flip-flops. Let the contents of the Flip-flops be denoted by X , Y , Z and W . Given that the function $f(X, Y, Z, W) = Z \oplus W$, determine the State Equations and the State Table for the circuit. Assuming that the initial state is $XYZW = 1100$, determine the output sequence.

